A

Mini Project Report

on

LendLoop – Campus Resource Sharing System with Borrowing & Lending Services

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science And Engineering (Data Science)

by

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CERTIFICATE

This to certify that the Mini Project report on LendLoop campus resource sharing system with borrowing and lending has been submitted by Vedant Dumbare (24107129), Krutika Ghadi (24107079), Tejas Shelar (24107058) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science And Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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Chapter 1

Introduction

The rapid growth of digital technologies has significantly transformed how resources are accessed and shared in everyday life. In educational institutions, especially college campuses, students frequently require items such as books, calculators, lab equipment, and other academic resources for short durations. However, despite the availability of these resources within the campus, there is no structured system that enables efficient sharing among students.

Currently, students rely on informal methods such as personal contacts, messaging groups, or last-minute arrangements to borrow items. These methods are often unreliable, time-consuming, and lack transparency. As a result, students either fail to obtain the required resources on time or end up purchasing new items unnecessarily, leading to increased expenses and wastage of resources.

To address these challenges, **LendLoop – Campus Resource Sharing System** is introduced as a mobile-based application that provides a centralized platform for borrowing and lending resources within a college campus. The system allows users to list items, browse available resources, send borrowing requests, and manage transactions efficiently.

The application ensures secure access through user authentication and maintains all data in a centralized database, enabling real-time updates and consistency. By digitizing the process of resource sharing, LendLoop improves accessibility, reduces dependency on external platforms, and promotes a culture of collaboration among students.

1.1 Purpose:

The primary purpose of the LendLoop system is to develop a **centralized and user-friendly platform** that facilitates efficient sharing of resources within a college campus.

In traditional systems, borrowing and lending activities are unorganized and depend heavily on personal communication, which often leads to delays and inconvenience. LendLoop aims to eliminate these issues by providing a structured digital solution where students can easily find and share resources.

The system is designed to:

- Provide a single platform for all resource-sharing activities
- Simplify the process of borrowing and lending
- Ensure secure access through authentication
- Maintain transparency in request handling

Additionally, this project serves as a **practical implementation of modern mobile application development**, helping users understand system functionality and developers understand system design and architecture.

1.2 Problem statement:

In a college campus environment, students frequently require academic resources such as books, calculators, and lab equipment for short durations. However, due to the absence of a structured and centralized system, the process of borrowing and lending these resources is currently carried out through informal methods such as personal communication, messaing groups, or last-minute arrangements. These approaches are unorganized, unreliable, and lack transparency, making it difficult for students to access required items efficiently.

Although external platforms such as OLX or rental services are available, they are not suitable for campus usage as they are not restricted to verified users, focus mainly on buying and selling rather than sharing, and do not provide quick accessibility within a limited campus environment. This results in issues such as lack of trust, absence of proper verification, and difficulty in ensuring secure transactions.

Furthermore, the current system lacks proper tracking and management mechanisms. There is no way to monitor requests, approvals, or item availability, leading to confusion, misuse, and lack of accountability. Due to these limitations, students often fail to obtain necessary resources on time or are forced to purchase new items, resulting in increased expenses and inefficient utilization of available resources.

To overcome these challenges, there is a need for a **secure, centralized, and campus specific system** that enables efficient borrowing and lending of resources. The proposed system, LendLoop, provides a mobile-based platform where users can list items, browse available resources, send and manage requests, and track transactions in real time. By integrat-

ing user authentication and centralized data management, the system ensures security, transparency, and ease of access, thereby improving resource utilization and promoting collaboration within the campus.

1.3 Objectives

The LendLoop system is developed with the following objectives:

1. To Develop a Centralized Resource Sharing Platform

A unified and centralized platform is established to enable students to list, search, and access resources within the campus efficiently. In traditional environments, resource sharing is scattered across informal channels such as personal contacts or messaging groups, which leads to inefficiency and lack of accessibility.

By bringing all resources into a single digital platform, the system ensures that users can easily view available items without relying on manual communication. This centralization improves visibility of resources, reduces time spent searching for items, and ensures better organization. It also creates a structured environment where all activities related to borrowing and lending are managed in one place, thereby enhancing usability and reliability

2. To Develop Efficient Borrowing and Lending Process

A structured workflow is implemented to manage the borrowing and lending of items in a systematic manner. The system allows users to send borrowing requests, which are then reviewed by item owners who can approve or reject them based on availability and preference.

This approach eliminates confusion and miscommunication that commonly occurs in informal systems. Each transaction is clearly recorded and tracked, ensuring transparency throughout the process. The request-based mechanism also ensures that item owners retain full control over their resources while borrowers are provided with a clear and reliable method to access items.

Overall, this objective focuses on improving efficiency, reducing delays, and ensuring smooth interaction between users.

3. To Develop Secure User Authentication

Security and trust are essential for any resource-sharing system. The platform ensures that only authorized users can access the system through Firebase Authentication. This authentication mechanism verifies user identity using secure login credentials, preventing unauthorized access.

By restricting access to verified students within the campus, the system creates a trusted environment for sharing resources. This reduces the risk of misuse, fraud, or unauthorized transactions. Additionally, secure authentication enhances accountability, as all actions performed within the system are associated with verified users.

This objective plays a crucial role in maintaining system integrity and ensuring safe interaction among users.

4. Real-Time Data Management

Efficient data handling is ensured through the use of Firebase Realtime Database, which allows all system data to be stored and updated dynamically. Any changes made within the application, such as adding items, sending requests, or updating request status, are instantly reflected across all users. This real-time synchronization eliminates delays and ensures that users always have access to the most recent and accurate information. It also improves system responsiveness and reliability, providing a smooth and seamless user experience while maintaining data consistency throughout the platform.

1.4 Scope

The scope of the LendLoop system focuses on providing a centralized platform for resource sharing within a campus. The system allows users to list items, search for available resources, and manage borrowing requests in a structured manner. It improves accessibility, communication, and transparency compared to traditional informal methods.

The application uses Firebase for real-time data management and secure user authentication, ensuring efficient and reliable operation. The system is designed to be scalable and can be enhanced in the future with features such as notifications, advanced search, and multi-campus support. However, the current implementation is limited to basic resource sharing within a campus environment.

Chapter 2

Proposed system

In order to overcome the limitations of the existing system, a structured and technology-driven approach is required to ensure efficient resource sharing within a campus environment. The proposed system focuses on providing a **centralized, secure, and real-time platform** that enables users to manage borrowing and lending activities in a systematic manner.

The system is designed using a mobile application interface integrated with a cloud-based backend, ensuring seamless interaction between users and data storage. It follows a modular architecture where different components handle specific functionalities such as user authentication, item management, request handling, and database operations.

The overall working of the system can be understood through the block diagram shown in Figure 2.1

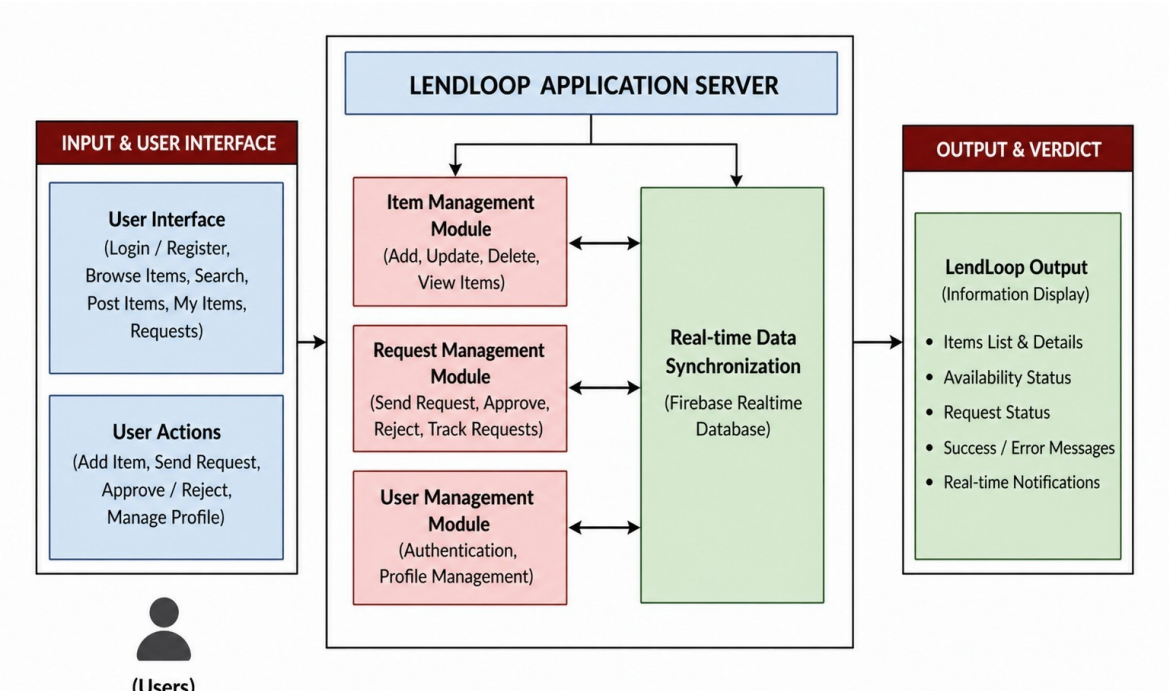

Figure 2.1.Block Diagram

As shown in the Figure 2.1 the LendLoop system is shown in the above block diagram. It represents the flow of data and interaction between different components of the system, starting from the user and ending at the database and supporting services.

At the top level, the system begins with **users**, who can act as both lenders and borrowers. These users interact with the system through the **mobile application**, which serves as the main interface. The mobile app provides various functionalities such as login/registration, dashboard access, searching items, posting items, viewing requests, and managing user profiles.

Once the user performs any action, the request is passed to the **authentication module**, which uses Firebase Authentication to verify user identity. This ensures that only authorized users can access the system and perform operations securely.

After successful authentication, the request is processed by the **application logic layer**. This layer is responsible for handling core functionalities of the system, including user management, item management, search and filtering of items, request handling (send, approve, reject), notifications, and maintaining transaction history. It acts as the brain of the system where all decision-making and processing takes place.

The processed data is then stored and managed in the **Firebase Realtime Database**, which acts as the central data storage. It contains multiple collections such as users, items, requests, chats/notifications, and transaction history. This database ensures that all data is stored in a structured and organized manner.

The system also integrates additional supporting components. The **real-time sync module** ensures that any updates in the database are instantly reflected across all users. The **push notification service** informs users about request updates, approvals, and reminders. The **file storage service** (Firebase Storage) is used to store images related to items. Additionally, the **logging and analytics module** helps in tracking system usage and maintaining error logs.

Overall, the architecture provides a **centralized, secure, and real-time system** that ensures efficient communication between users and seamless management of borrowing and lending activities within the campus.

2.1 Features and Functionality

The proposed system provides a set of integrated features that ensure efficient, secure, and user-friendly resource sharing within the campus. These features are designed to handle the complete workflow of borrowing and lending activities.

1. User Authentication

The system provides a secure login and registration mechanism using Firebase Authentication.

Only verified users can access the application, ensuring trust and preventing unauthorized usage.

2. Item Listing and Management

Users can add items they want to lend by providing details such as item name, category, and description.

The system also allows users to update or delete their listed items, ensuring proper management of resources.

3. Search and Browse Items

The application enables users to browse available items within the campus.

Users can search for specific resources, making it easy to find required items quickly.

4. Borrow Request System

Users can send borrowing requests for items listed by other users.

This feature provides a structured and systematic way to request resources instead of relying on informal methods.

5. Request Management (Approve / Reject)

Item owners can view incoming requests and take actions such as approving or rejecting them.

This ensures that the owner has full control over their items.

6. Request Tracking

Users can track the status of their requests (Pending, Approved, Rejected).

This improves transparency and keeps users informed about their transactions.

7. Real-Time Data Management

The system uses Firebase Realtime Database to store and manage all data.

Any changes in data are updated instantly across all users, ensuring consistency and accuracy.

8. Notification System

The system provides notifications for important events such as request approval or rejection.

This keeps users updated without the need to manually check the application.

9. CRUD Operations

The system supports full CRUD functionality:

- Create: Add new items
- Read: View available items
- Update: Modify item details
- Delete: Remove items

Chapter 3

Project Outcome

The LendLoop – Campus Resource Sharing System has been successfully designed and implemented to provide a centralized solution for resource sharing within a college campus. The system effectively addresses the challenges associated with informal borrowing methods and lack of structured resource management.

The project achieves its primary objective by offering a platform where students can easily list, browse, and borrow items in a secure and organized manner. The implementation of Firebase Authentication ensures that only verified users can access the system, thereby improving trust and preventing misuse.

The use of Firebase Realtime Database enables efficient data handling and real-time synchronization of information across all users. This ensures that item availability, request status, and user interactions are updated instantly, providing a smooth and reliable user experience.

The system successfully simplifies the process of borrowing and lending by introducing a structured workflow that includes request submission, approval or rejection by item owners, and tracking of request status. This enhances transparency and accountability in all transactions.

Additionally, the platform promotes efficient utilization of available resources by reducing unnecessary purchases and encouraging sharing among students. It also fosters collaboration within the campus by creating a trusted environment for resource exchange.

Overall, the system meets all the defined objectives and proves to be a practical, efficient, and scalable solution for campus resource sharing. It demonstrates the effective use of modern technologies in solving real-world problems and provides a strong foundation for future enhancements.

Chapter 4

Software Requirements

The software requirements define the tools, technologies, and platforms used for the development and execution of the LendLoop system. These technologies ensure that the application is efficient, scalable, and capable of handling real-time operations.

4.1 Development Environment

The application is developed using Android Studio, which is the official Integrated Development Environment (IDE) for Android application development.

Android Studio provides a comprehensive environment that includes:

- Code editor with syntax highlighting and auto-completion
- Debugging and performance analysis tools
- Emulator for testing applications on virtual devices
- Gradle-based build system for dependency management

It supports modern development practices and ensures efficient coding, testing, and deployment of the application.

4.2 Programming Language

The system is developed using Java / Kotlin, which are the officially supported languages for Android development.

- Java: Used for implementing core application logic, object-oriented programming, and backend communication.
- Kotlin: A modern, concise, and null-safe programming language that improves code readability and reduces boilerplate code.

Java Version:

The application is developed using Java Development Kit (JDK) 8 or above, which provides features such as:

- Lambda expressions
- Improved memory management
- Enhanced performance

Both languages ensure high performance, reliability, and compatibility with Android SDK.

4.3 Backend and Database (Firebase)

The backend of the system is implemented using Firebase, a cloud-based platform provided by Google that supports real-time application development.

Firebase Services Used:

1. Firebase Authentication

Provides secure user authentication using email and password.

Ensures that only verified users can access the system.

2. Firebase Realtime Database

A NoSQL cloud database used to store:

- User data
- Item details
- Borrowing requests

It enables real-time data synchronization across all users.

3. Firebase Storage

Used for storing images or files related to items.

Provides scalable and secure cloud storage.

4.4 Operating System

The application is designed to run on Android OS (version 7.0 and above), ensuring compatibility with a wide range of smartphones used by students.

4.5 Internet Requirement

A stable internet connection is required for:

- User authentication
- Data retrieval and updates
- Sending and receiving requests in real time

4.6 Tools and APIs

- Android SDK: Provides libraries and tools for building Android applications
- Firebase APIs: Used for authentication, database operations, and storage
- Gradle Build System: Manages dependencies and builds the application efficiently

Chapter 5

Project Design

The project design defines the overall structure and workflow of the system. It explains how different components interact with each other to perform various operations such as authentication, item listing, searching, and request handling.

The system follows a **modular and layered design approach**, where each module performs a specific function. This design improves maintainability, scalability, and efficiency of the application. The interaction between the user interface, application logic, and database ensures smooth and real-time operation of the system.

The workflow of the system can be better understood using diagrams that represent different user actions and system responses.

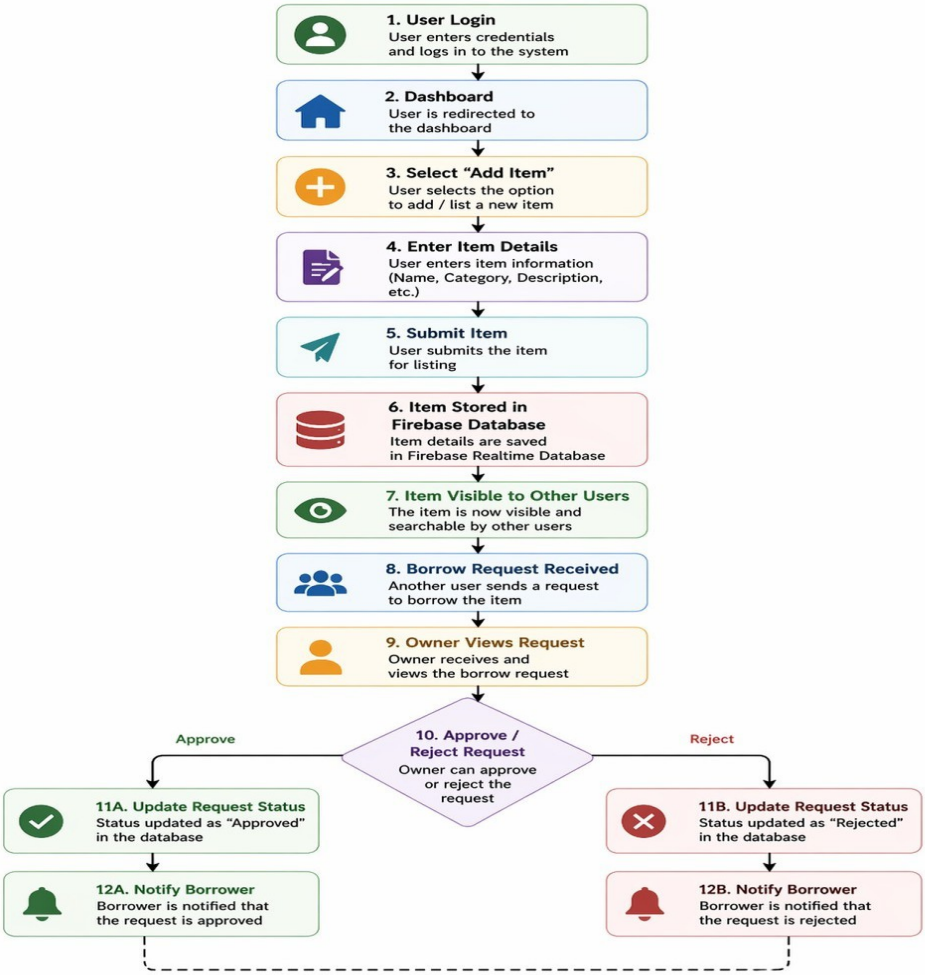

Figure 5.1:Workflow for lending an Item

The workflow for lending an item, as shown in Figure 5.1, represents the complete process through which a user lists an item and manages borrowing requests within the system. The process begins when the user opens the application and logs in using valid credentials. The authentication is handled securely, ensuring that only authorized users can access the system. After successful login, the user is redirected to the dashboard, which serves as the central interface providing access to all features of the application.

From the dashboard, the user selects the option to add or list a new item. This feature allows the user to contribute resources to the platform, making them available for other users. The system then prompts the user to enter detailed information about the item, including the item name, category, and description. These details are essential as they help other users understand the nature and condition of the item, enabling them to make informed decisions while requesting it.

Once the user enters all the required details, the item is submitted to the system. At this stage, the application processes the input and stores the item data in the Firebase Realtime Database. Since the database supports real-time synchronization, the newly added item becomes instantly visible to all other users of the application. This eliminates delays and ensures that the system always reflects the most up-to-date information.

After the item is listed, other users can browse the available items and send borrowing requests. When a request is sent, it is recorded in the database and linked to the respective item owner. The owner receives this request and can view it through the request management section of the application. This section provides details about the requester, allowing the owner to evaluate the request properly.

The item owner then reviews the incoming requests and makes a decision based on availability and preference. The system provides options to approve or reject each request. This step is crucial as it ensures that the owner retains full control over their resources. If the request is approved, the borrower is granted access to the item; if rejected, the request is declined.

Once the decision is made, the system updates the request status in the database. Due to real-time synchronization, this update is immediately reflected on the borrower's side. The borrower is also notified about the outcome of the request, ensuring clear communication between both parties.

This workflow ensures that the lending process is structured, transparent, and efficient. It eliminates the need for informal communication methods and provides a reliable mechanism for managing resources. By maintaining proper records of all transactions, the system also ensures accountability and reduces the chances of misuse.

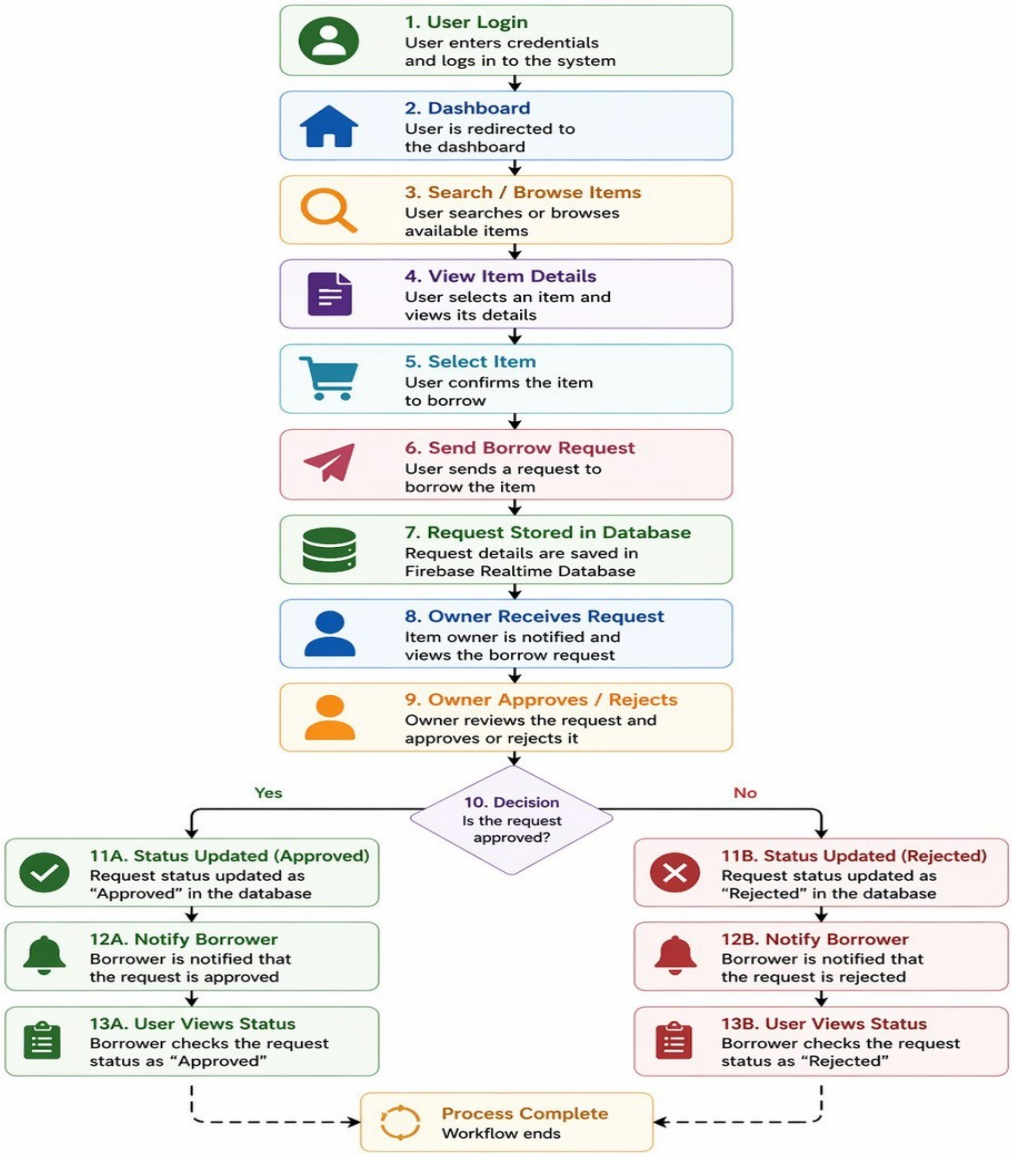

Figure 5.2:Workflow for searching and borrowing an item

The workflow for searching and borrowing an item, as shown in Figure 5.2, illustrates how a user interacts with the system to find and request resources efficiently. The process begins when the user logs into the application using valid credentials. The authentication mechanism ensures secure access and verifies that the user is authorized to use the system. After successful login, the user is directed to the dashboard, which acts as the main interface for accessing different features.

From the dashboard, the user can search or browse the list of available items. The system provides a structured and organized display of items, making it easy for users to locate the required resources. The search functionality allows users to enter keywords and quickly filter relevant items, improving efficiency and reducing the time required to find specific resources.

Once the user identifies an item of interest, they can view its detailed information. This includes the item description, category, and availability status. Providing detailed information helps users make informed decisions before sending a request. It also reduces the chances of confusion or incorrect requests.

After reviewing the item details, the user selects the item and proceeds to send a borrow request. This request is processed by the system and stored in the Firebase Realtime Database. The request contains all necessary information, including the borrower details and item information, ensuring proper tracking of transactions.

The item owner receives the request and is notified through the system. The owner can access the request management section to view all incoming requests. Each request is presented with sufficient details, allowing the owner to evaluate it effectively.

The owner then decides whether to approve or reject the request. This decision is important as it ensures that the owner maintains control over their items. If the request is approved, the borrower is allowed to proceed with borrowing the item. If rejected, the borrower must look for alternative options.

Once the owner takes action, the system updates the request status in real time. This update is instantly reflected in the borrower's application due to the real-time capabilities of the database. The borrower can check the updated status at any time, ensuring transparency in the process.

This workflow provides a systematic and efficient approach to searching and borrowing items. It reduces dependency on manual communication, improves accessibility, and ensures that all transactions are properly tracked and managed. By integrating real-time updates and structured request handling, the system enhances user experience and ensures smooth operation.

Chapter 6

Project Scheduling

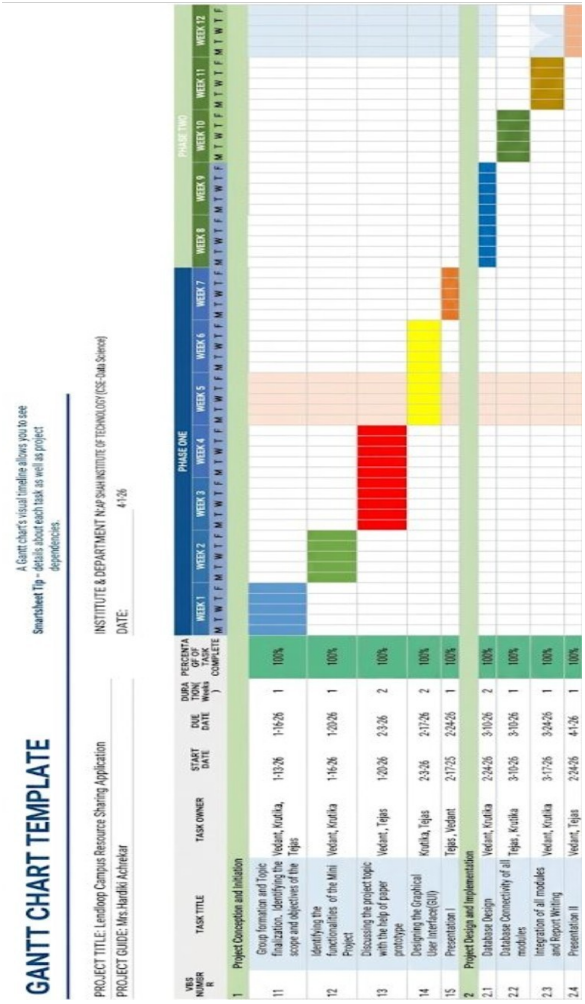

Figure 6.1:Gant Chart

Project scheduling is an essential part of software development that helps in planning and organizing tasks efficiently over a specific timeline. It ensures that each phase of the project is completed in a structured and systematic manner. In this project, the development process was divided into multiple phases such as problem identification, system design, implementation, testing, and final deployment. A Gantt chart was used to represent the timeline and track the progress of each phase. Proper scheduling helped in managing time effectively, avoiding delays, and ensuring the successful completion of the project within the given timeframe.

Chapter 7

Results

The developed system was successfully implemented and tested on an Android platform. The application provides a user-friendly interface and ensures smooth functionality for all major operations such as login, item listing, searching, borrowing, and request management.

The following sections present the key screens of the application along with their explanations

7.1 Login Page

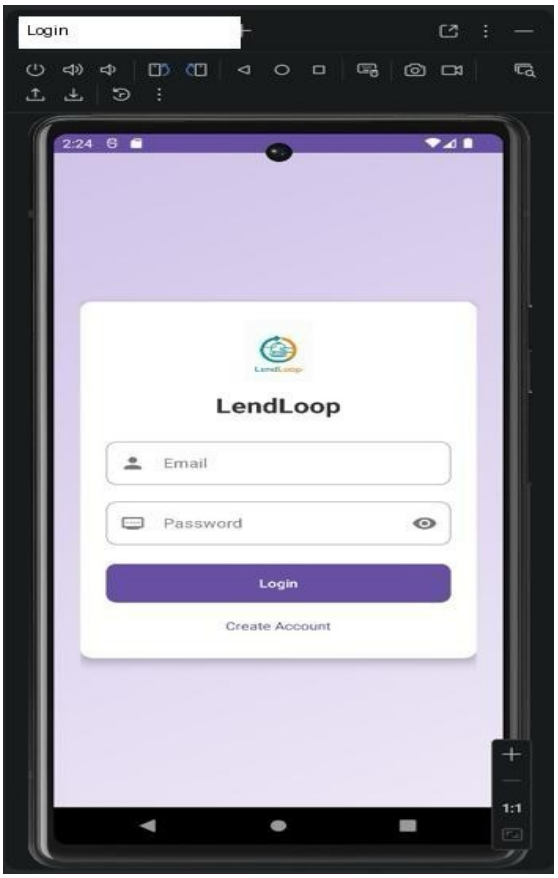

Figure 7.1: Login Page

The login page of the application is shown in Figure 7.1. This is the initial screen that appears when the user opens the application. It allows users to enter their registered email and password to access the system.

The authentication process is handled using Firebase Authentication, ensuring that only verified users can log in. The interface is designed to be simple and user-friendly, providing quick access to the system. Additionally, users who do not have an account can navigate to the registration page from this screen.

7.2 Dashboard

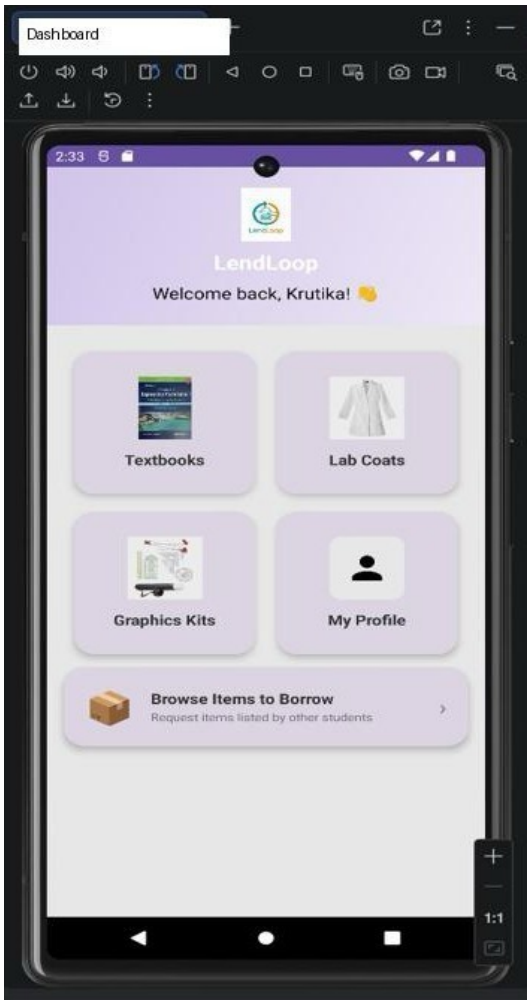

Figure 7.2: Dashboard

The dashboard is illustrated in Figure 7.2. After successful login, the user is redirected to this screen, which acts as the main interface of the application.

The dashboard provides access to key features such as browsing items, viewing categories (textbooks, lab equipment, etc.), and managing user profile. It allows users to navigate easily through different

functionalities of the system. The layout is organized using card-based design, improving usability and user experience.

7.3 Search and Browsing

Figure 7.3: Search and Browse

The search and browse functionality is shown in Figure 7.3. This screen allows users to search for available items within the campus.

Users can enter keywords in the search bar to find specific items. The system displays results in a structured format, making it easy to view item details and availability. This feature improves efficiency by reducing the time required to locate resources.

7.4 Post Items

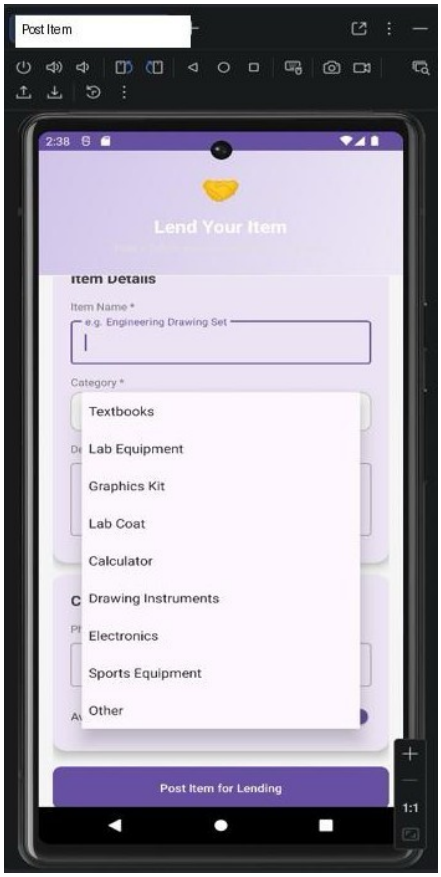

Figure 7.4: Post Items

The item posting interface is shown in Figure 7.4. This screen allows users to list items they wish to lend.

Users are required to enter item details such as name, category, and description. The system also provides a dropdown menu for selecting item categories. Once the details are submitted, the item is stored in the database and becomes visible to other users.

7.5 Request Management

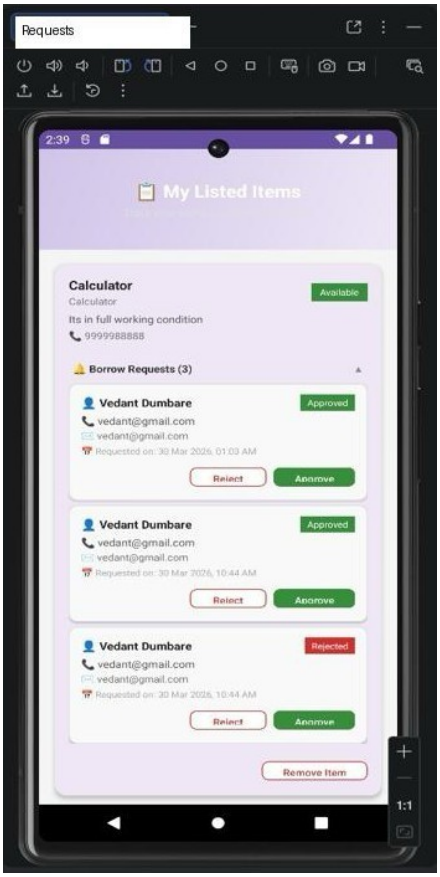

Figure 7.5:Request Management

The request management screen is illustrated in Figure 7.5. This screen allows item owners to manage incoming borrowing requests.

Users can view all requests related to their listed items and take appropriate actions such as approving or rejecting them. This feature ensures proper control over resources and maintains transparency in transactions.

Chapter 8

Conclusion

The LendLoop system has been successfully designed and implemented as a centralized platform for resource sharing within a campus environment. The main objective of the system was to simplify the process of borrowing and lending items among students by replacing informal and unstructured methods with a well-organized digital solution. The application provides a user-friendly interface through which users can list items, search for available resources, and manage borrowing requests efficiently.

The system effectively addresses key issues such as lack of accessibility, poor communication, and absence of proper tracking in traditional resource-sharing methods. By integrating all functionalities into a single platform, LendLoop ensures better visibility and organization of resources. Users can easily browse items, view their details, and send requests, while item owners can approve or reject these requests, ensuring transparency and control over their resources.

The use of Firebase as a backend plays an important role in maintaining real-time data synchronization and secure user authentication. All updates, such as item listings and request status changes, are instantly reflected across the system, improving reliability and enhancing user experience. The system also reduces manual effort, minimizes errors, and provides a scalable structure for future enhancements.

The results of the system demonstrate that it successfully meets its intended objectives by providing an efficient, reliable, and practical solution for campus resource sharing. It promotes optimal utilization of resources, reduces unnecessary expenses for students, and encourages collaboration within the campus community.

Furthermore, the system has strong potential for future improvements and expansion. It can be enhanced by integrating online payment features for renting items, enabling secure transactions within the platform. The addition of real-time notifications and alerts can improve communication by informing users about request updates, approvals, and reminders. A rating and review system can also be introduced to build trust among users and improve reliability.

In addition, advanced search capabilities can be implemented to enhance item discovery and improve usability. The system can be expanded to support multiple campuses or institutions, making it scalable for a larger user base. Features such as in-app chat can facilitate direct communication between users, improving coordination during borrowing and lending processes.

Further enhancements may include data analytics to track usage patterns and improve system performance, along with deployment on multiple platforms such as web and iOS to increase accessibility. Security features can also be strengthened to ensure better protection of user data and privacy.

Overall, the LendLoop system provides a strong foundation for a modern resource-sharing platform and has the potential to evolve into a more advanced and comprehensive solution with the integration of additional features and technologies.

Chapter 9

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